

CSA0925 PROGRAMMING IN JAVA

CO1 AT-1

Q1. Design a Class Hierarchy (Pseudocode)

PSUEDO CODE

START

CLASS Person

DATA:

 Name

 Age

METHOD getPerson()

 INPUT Name

 INPUT Age

END METHOD

END CLASS

CLASS Student INHERITS Person

DATA:

 RegisterNumber

 Marks[5]

 Average

METHOD getStudent()

 CALL getPerson()

 INPUT RegisterNumber

 FOR i ← 1 TO 5

 INPUT Marks[i]

 END FOR

END METHOD

METHOD calculateAverage()

 Sum ← 0

 FOR i ← 1 TO 5

 Sum ← Sum + Marks[i]

 END FOR

 Average ← Sum / 5

END METHOD

END CLASS

CLASS ResearchStudent INHERITS Student

DATA:

 ResearchArea

```

METHOD getResearchDetails()
    CALL getStudent()
    INPUT ResearchArea
END METHOD

METHOD displayCompleteDetails()
    PRINT "Name :", Name
    PRINT "Age :", Age
    PRINT "Register Number :", RegisterNumber

    PRINT "Marks :"
    FOR i ← 1 TO 5
        PRINT Marks[i]
    END FOR

    PRINT "Average :", Average
    PRINT "Research Area :", ResearchArea
END METHOD
END CLASS

MAIN
    CREATE object RS OF ResearchStudent

    CALL RS.getResearchDetails()
    CALL RS.calculateAverage()
    CALL RS.displayCompleteDetails()

END

```

Q2. Complete the Missing Pseudocode

Missing blanks:

```

FOR j ← i + 1 TO N - 1
    IF A[i] == A[j]
        DELETE A[j]
    ENDIF

```

ENTIRE PSUEDO CODE:

```

BEGIN

READ N
READ ARRAY A

FOR i ← 0 TO N-1
    FOR j ← i+1 TO N-1
        IF A[i] == A[j]
            DELETE A[j]
        ENDIF
    ENDFOR
ENDFOR

```

PRINT Modified Array

END

Q3. Debug the Pseudocode

Incorrect line:

RETURN length * breadth

Correct line:

METHOD area(length)
RETURN length * length

The second method remains:

METHOD area(length,breadth)
RETURN length * breadth

Q4. Trace the Execution

Pseudocode:

sum = 0

i=1 → odd → sum=-1

i=2 → even → sum=1

i=3 → odd → sum=-2

i=4 → even → sum=2

i=5 → odd → sum=-3

Output:

-3

Q5. Write Efficient Pseudocode

PSUEDO CODE:

BEGIN

READ N

READ ARRAY A

largest $\leftarrow -\infty$

secondLargest $\leftarrow -\infty$

FOR i $\leftarrow$ 0 TO N-1

IF A[i] > largest THEN

secondLargest $\leftarrow$ largest

largest $\leftarrow$ A[i]

ELSE IF A[i] > secondLargest AND A[i] $\neq$ largest THEN

secondLargest $\leftarrow$ A[i]

ENDIF

ENDFOR

PRINT secondLargest

END

Q6. Convert Problem Statement into Pseudocode (Interfaces)

PSUEDO CODE:

BEGIN

INTERFACE Payment

METHOD pay(amount)

END INTERFACE

CLASS CreditCard IMPLEMENTS Payment

METHOD pay(amount)

PRINT "Paid ", amount, " using Credit Card"

END METHOD

END CLASS

CLASS UPI IMPLEMENTS Payment

METHOD pay(amount)

```

        PRINT "Paid ", amount, " using UPI"
    END METHOD
END CLASS

CLASS NetBanking IMPLEMENTS Payment
    METHOD pay(amount)
        PRINT "Paid ", amount, " using Net Banking"
    END METHOD
END CLASS

MAIN

READ choice
READ amount

IF choice = 1
    p ← CreditCard
ELSE IF choice = 2
    p ← UPI
ELSE
    p ← NetBanking
ENDIF

CALL p.pay(amount)

END

```

Q7. Complete the Pseudocode

Pseudocode:

```

CLASS Animal

METHOD speak()
    PRINT "Animal"
END METHOD

END CLASS

CLASS Dog INHERITS Animal

METHOD speak()
    PRINT "Barks"
END METHOD

END CLASS

Animal obj ← Dog
obj.speak()

```

Output:

Barks

Q8. Find Logical Errors**Correct Constructor:**

CLASS Employee

DATA

id

name

CONSTRUCTOR(id,name)

 this.id ← id

 this.name ← name

END CONSTRUCTOR

END CLASS

Q9. Case Study (Integrated Problem)**PSUEDO CODE:**

BEGIN

CLASS Vehicle

DATA

vehicleNo

ownerName

METHOD getVehicle()

 INPUT vehicleNo

 INPUT ownerName

END METHOD

METHOD display()

 PRINT vehicleNo

 PRINT ownerName

END METHOD

END CLASS

CLASS Car INHERITS Vehicle

```

DATA
    fuelType

METHOD getCar()
    CALL getVehicle()
    INPUT fuelType
END METHOD

METHOD display()
    CALL Vehicle.display()
    PRINT fuelType
END METHOD

END CLASS

CLASS Truck INHERITS Vehicle

DATA
    loadCapacity

METHOD getTruck()
    CALL getVehicle()
    INPUT loadCapacity
END METHOD

METHOD display()
    CALL Vehicle.display()
    PRINT loadCapacity
END METHOD

END CLASS

MAIN

READ N

FOR i ← 1 TO N
    READ type

    IF type = "Car" THEN
        CREATE C AS Car
        CALL C.getCar()
        STORE C

    ELSE
        CREATE T AS Truck
        CALL T.getTruck()
        STORE T
    ENDIF

ENDFOR

FOR EACH Vehicle V
    CALL V.display()
ENDFOR

END

```

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